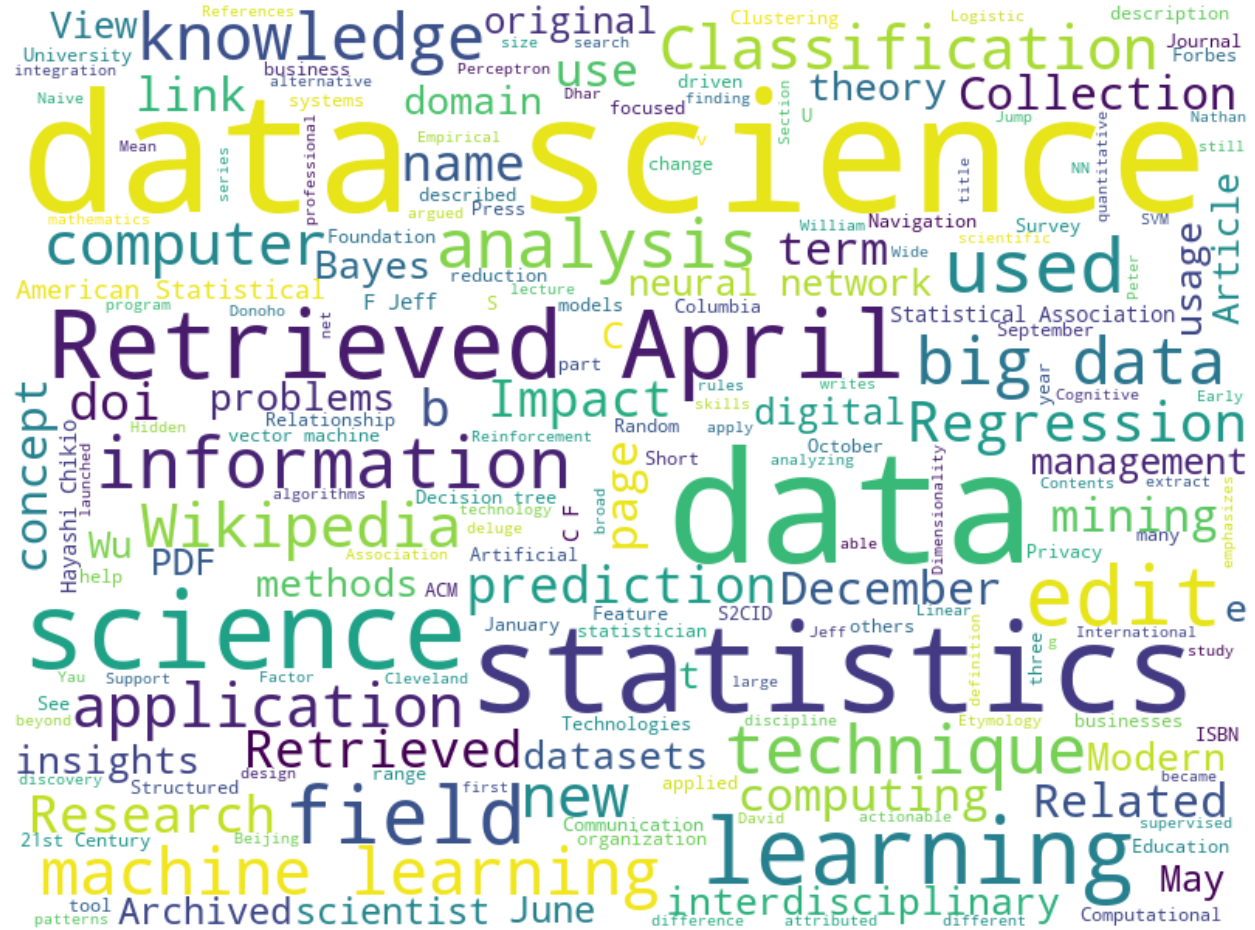
AL NAFI
Education benefits all

Data Science

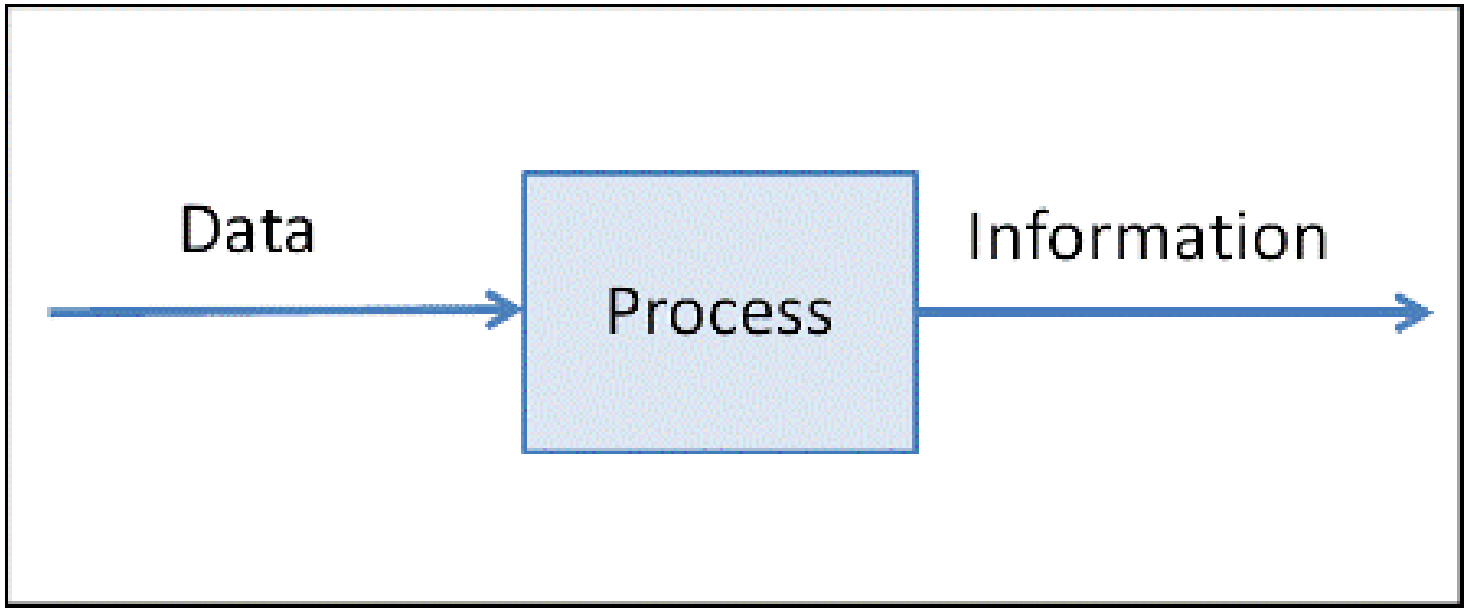


What is data?




data
science
knowledge
Classification
Collection
analysis
computer
Retrieved
April
big data
information
Regression
statistics
science
machine learning
learning
Wikipedia
prediction
application
technique
field
Research
Retrieved
insights
datasets
computing
Related
interdisciplinary
Archived
scientist
June
May
used
term
network
neural
Bayes
Impact
problems
doi
concept
digital
management
mining
edit
December
methods
PDF
Wu
Article
Journal
Forbes
theory
link
domain
use
original
name
Statistical Association
American Statistical
Cognitive
Education
Computational

we now use computers more and more for data processing and communication, rather than actual computations. When we write an e-mail to a friend or search for some information on the Internet - we are essentially **creating, storing, transmitting, and manipulating data.**



What is Data Science?



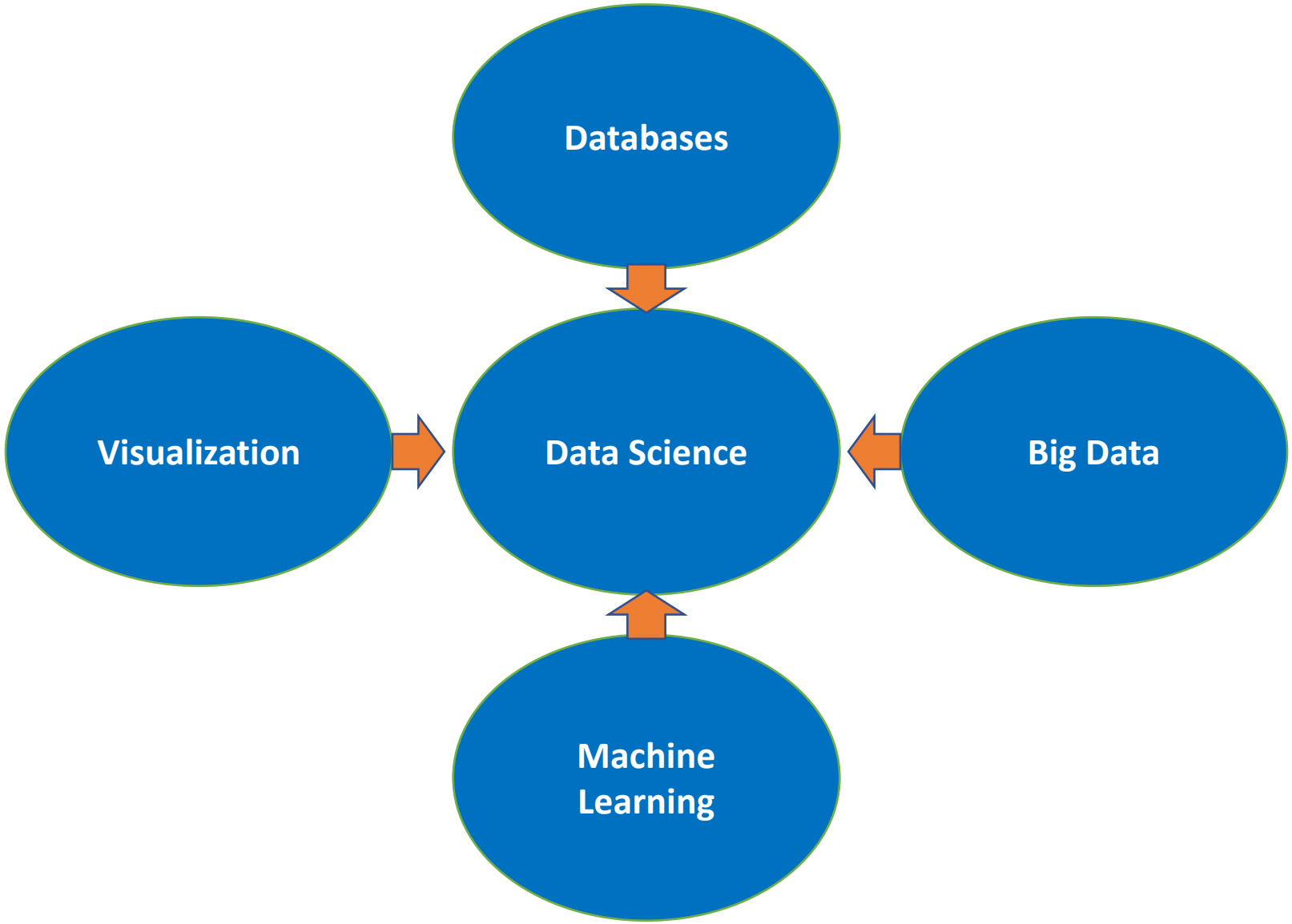
The main goal of data science is to **extract knowledge** from data, in other words - to **understand** data, find some hidden relationships and build a **model**.



Data science uses **scientific methods**, such as probability and statistics. In fact, when the term *data science* was first introduced, some people argued that data science was just a new fancy name for statistics. Nowadays it has become evident that the field is much broader.



Obtained knowledge should be applied to produce some **actionable insights**, i.e., practical insights that you can apply to real business situations.


Databases
Visualization
Data Science
Big Data
Machine Learning

Types of data

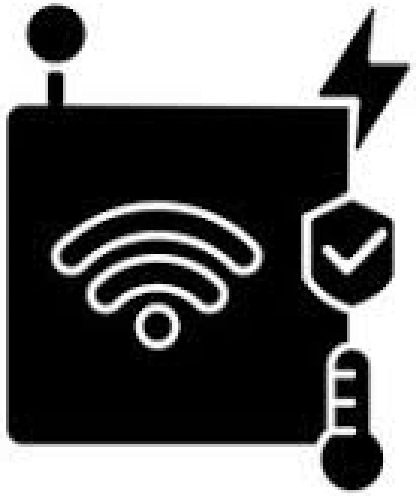
Structured	Semi-structured	Unstructured
List of people with their phone numbers	Wikipedia pages with links	Text of Encyclopedia Britannica
Temperature in all rooms of a building at every minute for the last 20 years	Collection of scientific papers in JSON format with authors, data of publication, and abstract	File share with corporate documents
Data for age and gender of all people entering the building	Internet pages	Raw video feed from surveillance camera

Where to get Data

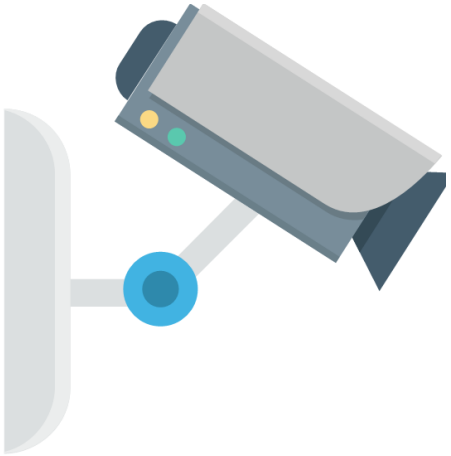












What you can do with Data



DATA ACQUISITION



DATA STORAGE



DATA PROCESSING



VISUALIZATION



TRAINING A PREDICTIVE MODEL

What you will learn in this course






















```python
def dijkstra(graph,src,dest,visited=[],distances={},predecessors={}):
    """ calculates a shortest path tree routed in src
    """
    if src not in graph:
        raise TypeError('The root of the shortest path tree cannot be found')
    if dest not in graph:
        raise TypeError('The target of the shortest path cannot be found')
    if src == dest:
        path=[]
        pred=dest
        while pred != None:
            path.append(pred)
            pred=predecessors.get(pred,None)
            readable=path[0]
        for index in range(1,len(path)): readable = path[index]+'--->'+readable
        print('shortest path - array: '+str(path))
        print("path: "+readable+",   cost="+str(distances[dest]))
    else :
        if not visited:
            distances[src]=0
        for neighbor in graph[src] :
            if neighbor not in visited:
                new_distance = distances[src] + graph[src][neighbor]
                if new_distance < distances.get(neighbor,float('inf')):
                    distances[neighbor] = new_distance
                    predecessors[neighbor] = src
        visited.append(src)
        unvisited={}
        for k in graph:
            if k not in visited:
                unvisited[k] = distances.get(k,float('inf'))
        x=min(unvisited, key=unvisited.get)
        dijkstra(graph,x,dest,visited,distances,predecessors)
if __name__ == "__main__":
    graph = {'A1': {'B1': 1, 'B2': 1.41, 'A2': 1},
             'A2': {'B1': 1.41, 'B2': 1, 'B3': 1.41, 'A1': 1, 'A3': 1},
    ..........................................................
    dijkstra(graph,'A1','H9')
    shortest path - array: ['H9', 'G8', 'F8', 'E7', 'D6', 'D5', 'C4', 'C3', 'B2', 'A1']
path: A1--->B2--->C3--->C4--->D5--->D6--->E7--->F8--->G8--->H9,   cost=11.459999999999999
```

